



COMSATS University Islamabad

Department of Computer Science



Course Code: CSC262



Course Title: Artificial Intelligence Credit Hours: 3(2,1)



Lecture Hours/Week: 2



Lab Hours/Week: 3



Pre-Requisites: None

Why Use Google Colab?

+
•
0

Free access to
computing resources

Easy sharing and
collaboration

Ideal for Machine
Learning & Data
Science

Create/Upload/Share
notebooks

Import/Save
notebooks from/to
Google Drive

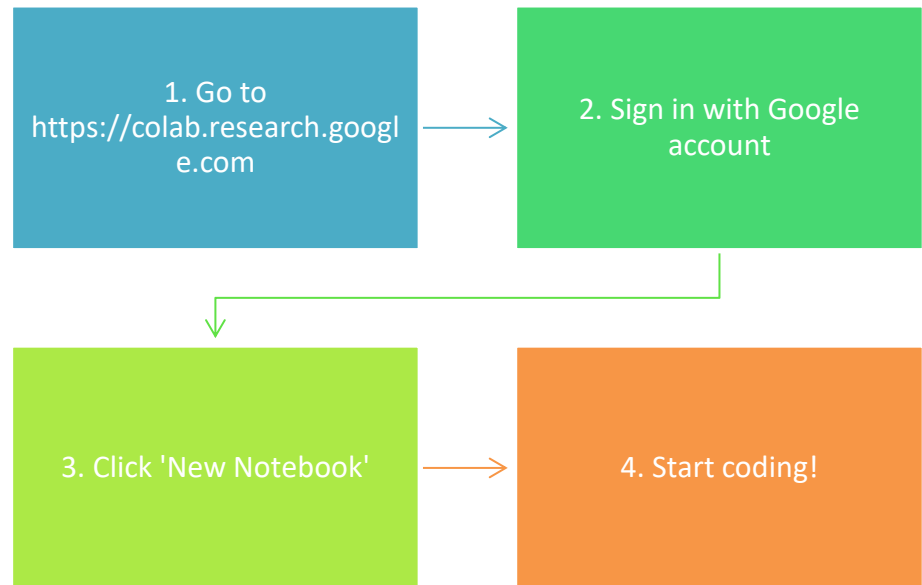
Import/Publish
notebooks from
GitHub

Import external
datasets e.g. from
Kaggle

Integrate PyTorch,
TensorFlow, Keras,
OpenCV

Free Cloud service
with free GPU

Getting Started



Colab Interface Overview

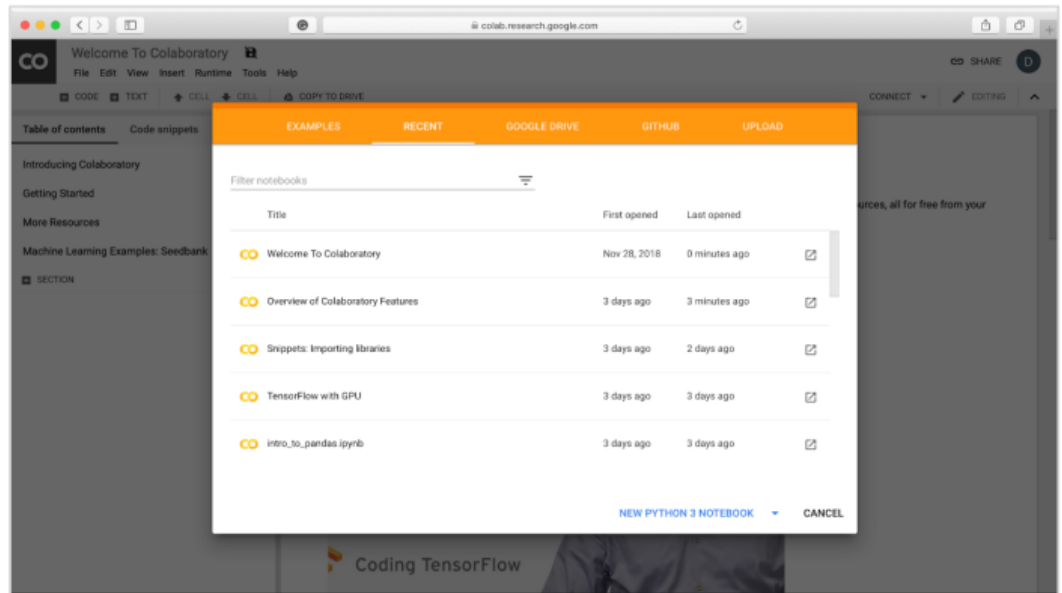
Code cells: Write Python code

Text cells: Add explanations (Markdown)

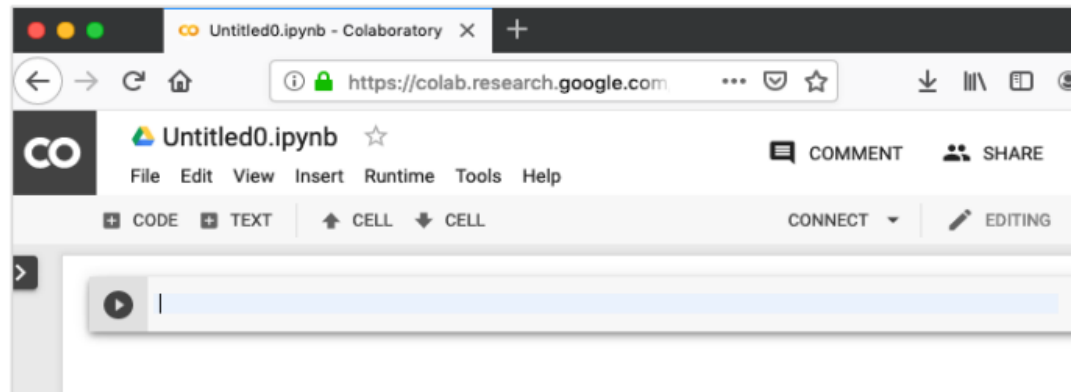
Run button: Execute code

Runtime menu: Manage resources

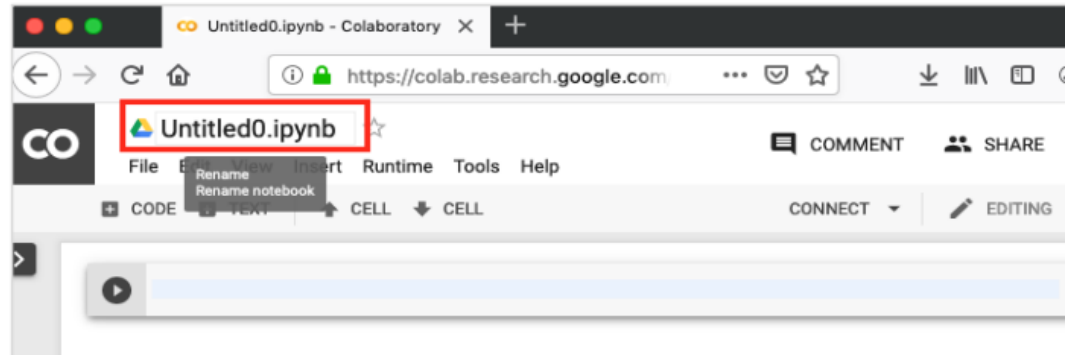
Your browser
would display
the following
screen
(assuming that
you are logged
into your
Google Drive):



Click on the NEW
PYTHON 3
NOTEBOOK link at
the bottom of the
screen. A new
notebook would
open up as shown
in the screen
below.

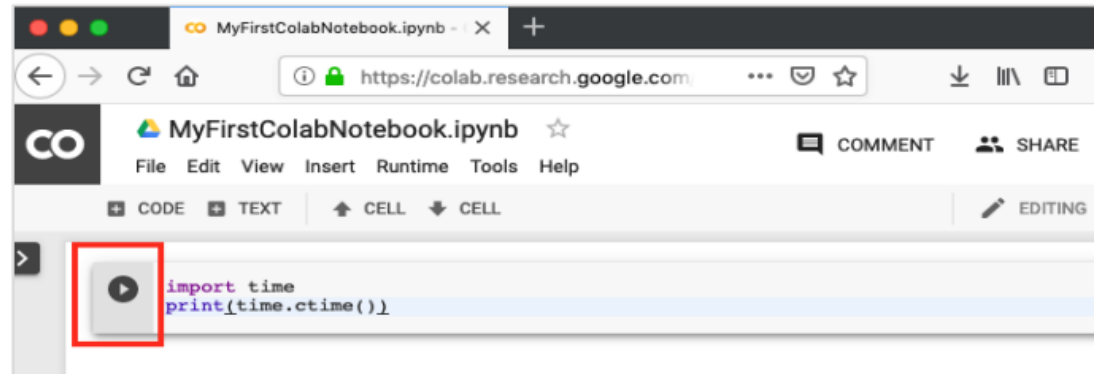


By default, the notebook uses the naming convention UntitledXX.ipynb. To rename the notebook, click on this name and type in the desired name in the edit box as shown here:

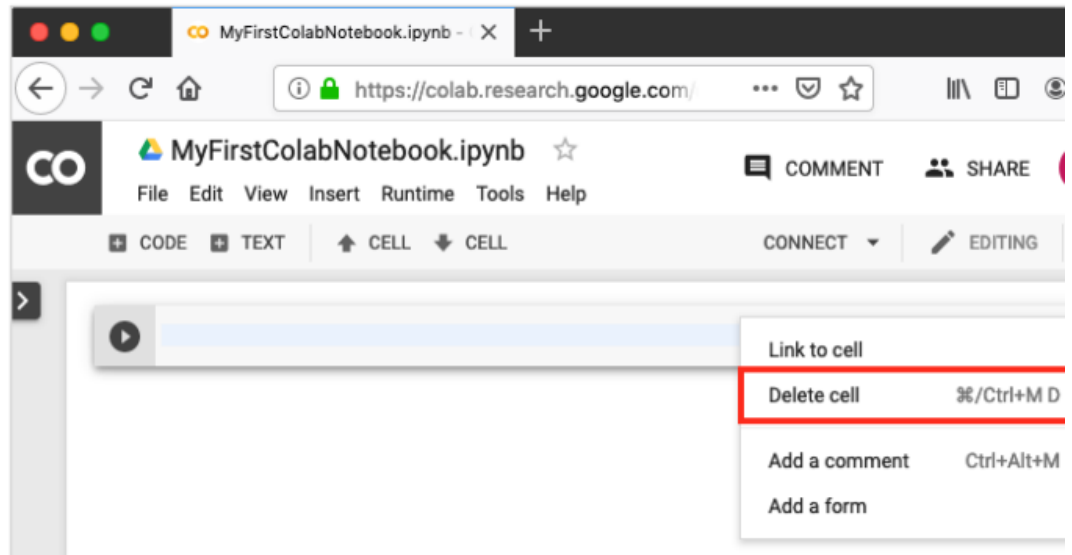


Entering Code

- You will now enter a trivial Python code in the code window and execute it. Enter the following two Python statements in the code window:
- `import time`
- `print(time.ctime())`



Deleting Cell



Google Colab – Documenting Your Code

- A text cell containing few mathematical equations typically used in ML is shown in the screenshot below:

Constraints are

- $3x_1 + 6x_2 + x_3 \leq 28$
- $7x_1 + 3x_2 + 2x_3 \leq 37$
- $4x_1 + 5x_2 + 2x_3 \leq 19$
- $x_1, x_2, x_3 \geq 0$

The trial vector is calculated as follows:

- $u_i(t) = x_i(t) + \beta(\hat{x}(t) - x_i(t)) + \beta \sum_{k=1}^{n_v} (x_{i1,k}(t) - x_{i2,k}(t))$

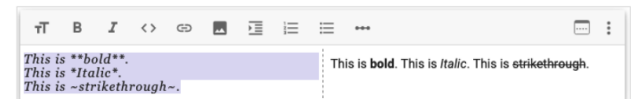
$$f(x_1, x_2) = 20 + e - 20\exp(-0.2\sqrt{\frac{1}{n}(x_1^2 + x_2^2)}) - \exp(\frac{1}{n}(\cos(2\pi x_1) + \cos(2\pi x_2))) \quad x \in [-5, 5]$$

$$A_{m,n} = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{pmatrix}$$

Markdown Examples

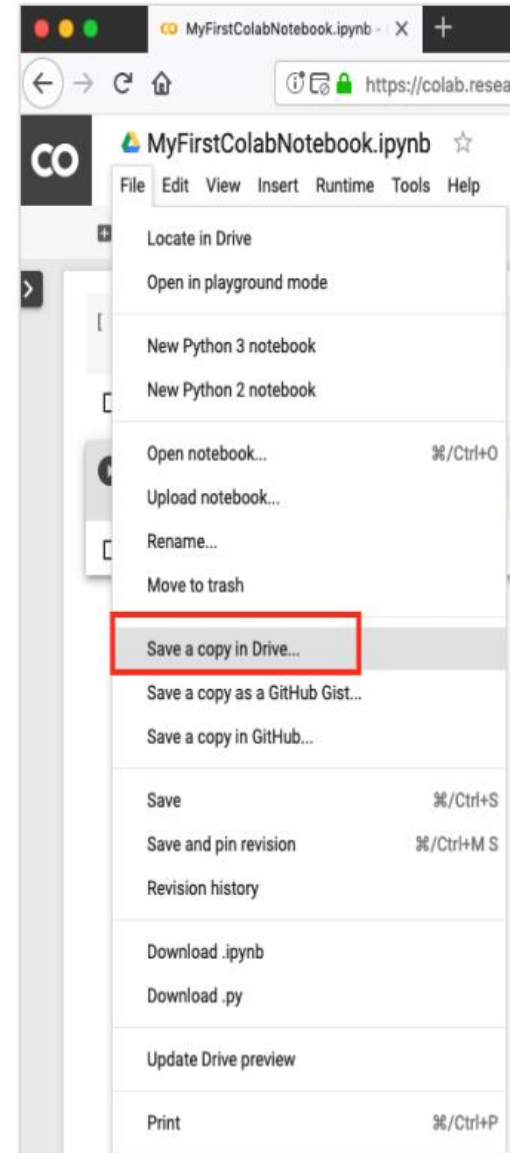
- Let us look into few examples of markup language syntax to demonstrate its capabilities. Type in the following text in the Text cell

```
This is **bold**.  
This is *italic*.  
This is ~strikethrough~.
```



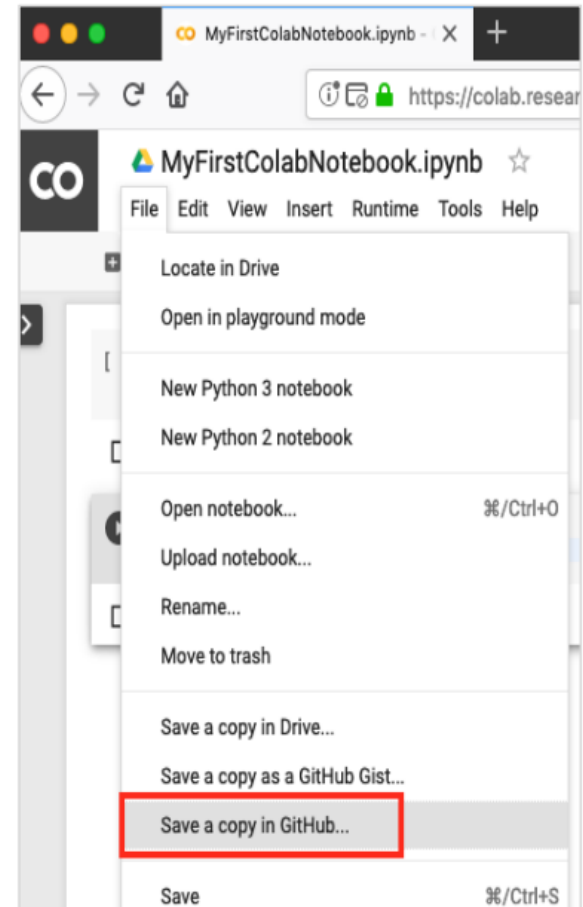
5. Google Colab – Saving Your Work

- Colab allows you to save your work to your Google Drive. To save your notebook, select the following menu options:

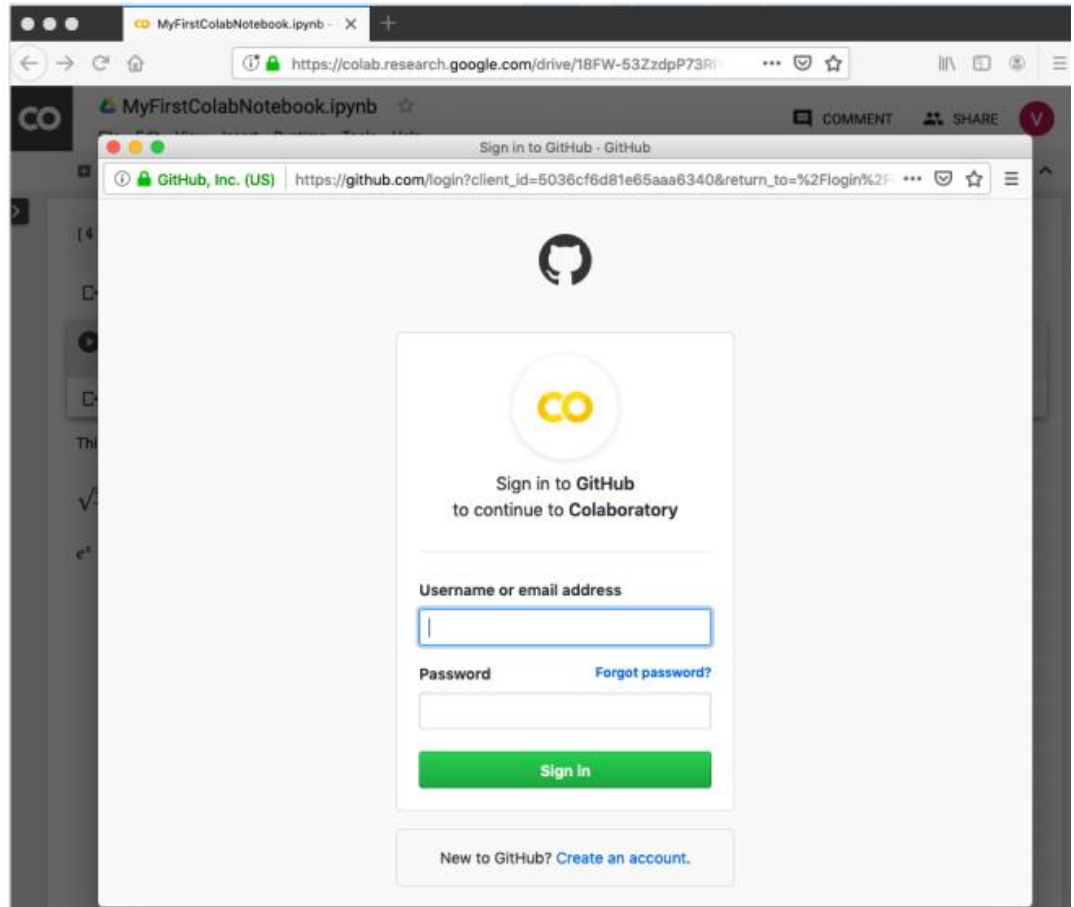


Saving to GitHub

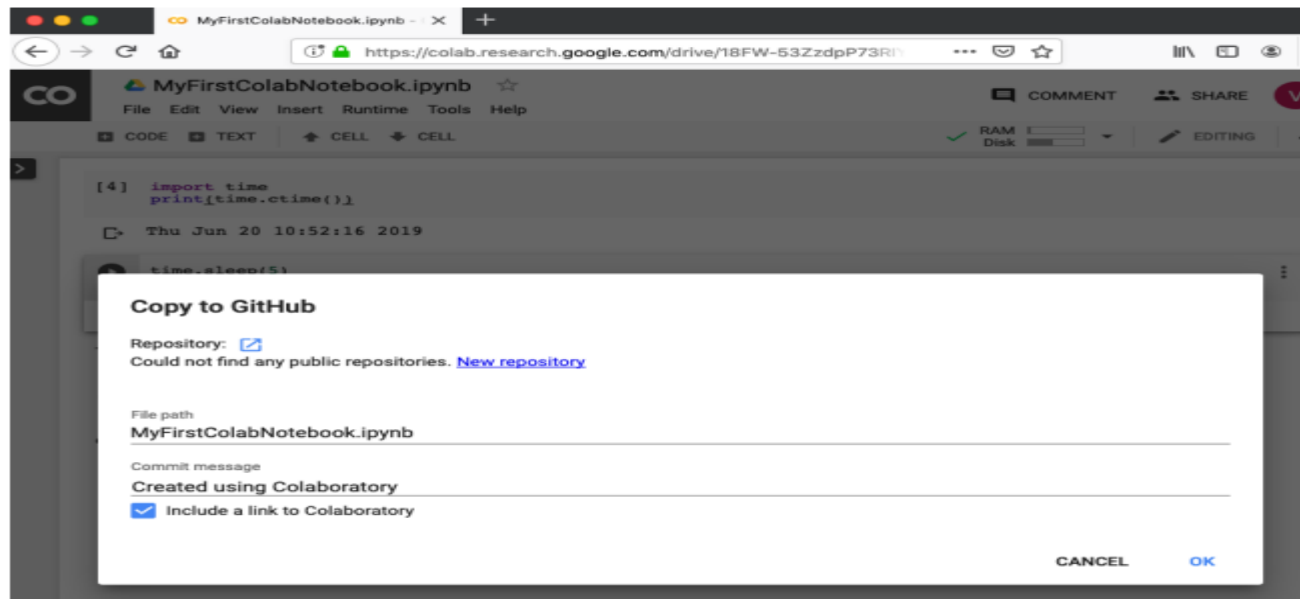
The menu selection is shown in the following screenshot for your quick reference:



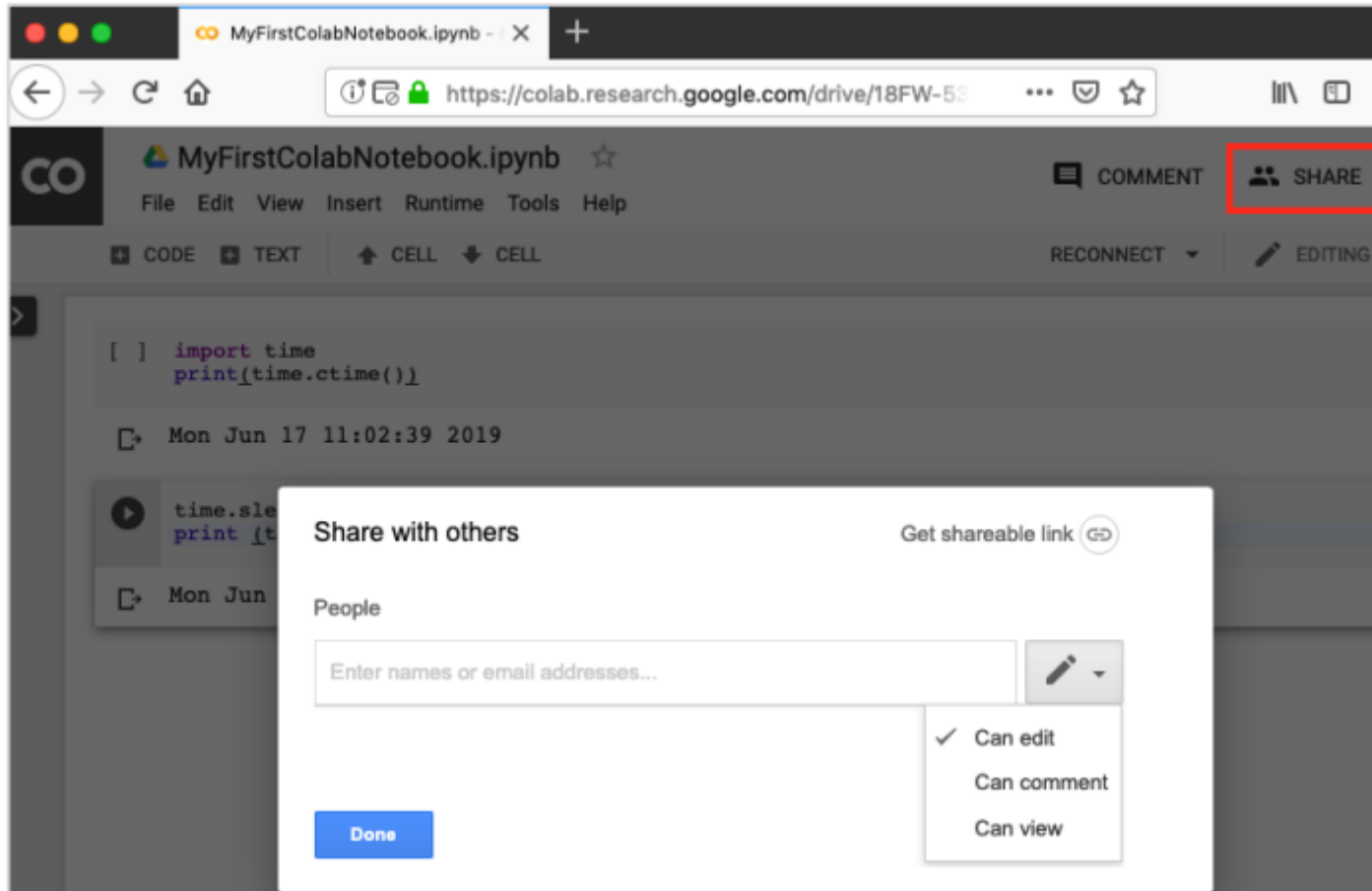
You will have
to wait until
you see the
login screen
to GitHub.



- Now, enter your credentials. If you do not have a repository, create a new one and save your project as shown in the screenshot below:



Google Colab – Sharing Notebook



Mounting Drive

```
# Run this cell to mount your Google Drive.  
from google.colab import drive  
drive.mount('/content/drive')
```

- If you run this code, you will be asked to enter the authentication code. The corresponding screen looks as shown below:



The screenshot shows a Google Colab interface. At the top, there is a code cell with a play button icon on the left and a three-dot menu icon on the right. The code inside the cell is: `# Run this cell to mount your Google Drive.`, `from google.colab import drive`, and `drive.mount('/content/drive')`. Below the code cell, there is a text prompt: "... Go to this URL in a browser: https://accounts.google.com/o/oauth2/auth?client_id=947318989803-6bn6qk8gdqf4n4q3pfee6491hcC". Below the URL, there is a text prompt: "Enter your authorization code:" followed by a text input field.

```
# Run this cell to mount your Google Drive.  
from google.colab import drive  
drive.mount('/content/drive')
```

... Go to this URL in a browser: https://accounts.google.com/o/oauth2/auth?client_id=947318989803-6bn6qk8gdqf4n4q3pfee6491hcC

Enter your authorization code:

Google Drive File System

This will allow **Google Drive File Stream** to:

-  See, edit, create, and delete all of your Google Drive files 
-  View the photos, videos and albums in your Google Photos 
-  View Google people information such as profiles and contacts 
-  See, edit, create, and delete any of your Google Drive documents 

Make sure you trust Google Drive File Stream

You may be sharing sensitive info with this site or app.
Learn about how Google Drive File Stream will handle your data by reviewing its terms of service and privacy policies.
You can always see or remove access in your [Google Account](#).

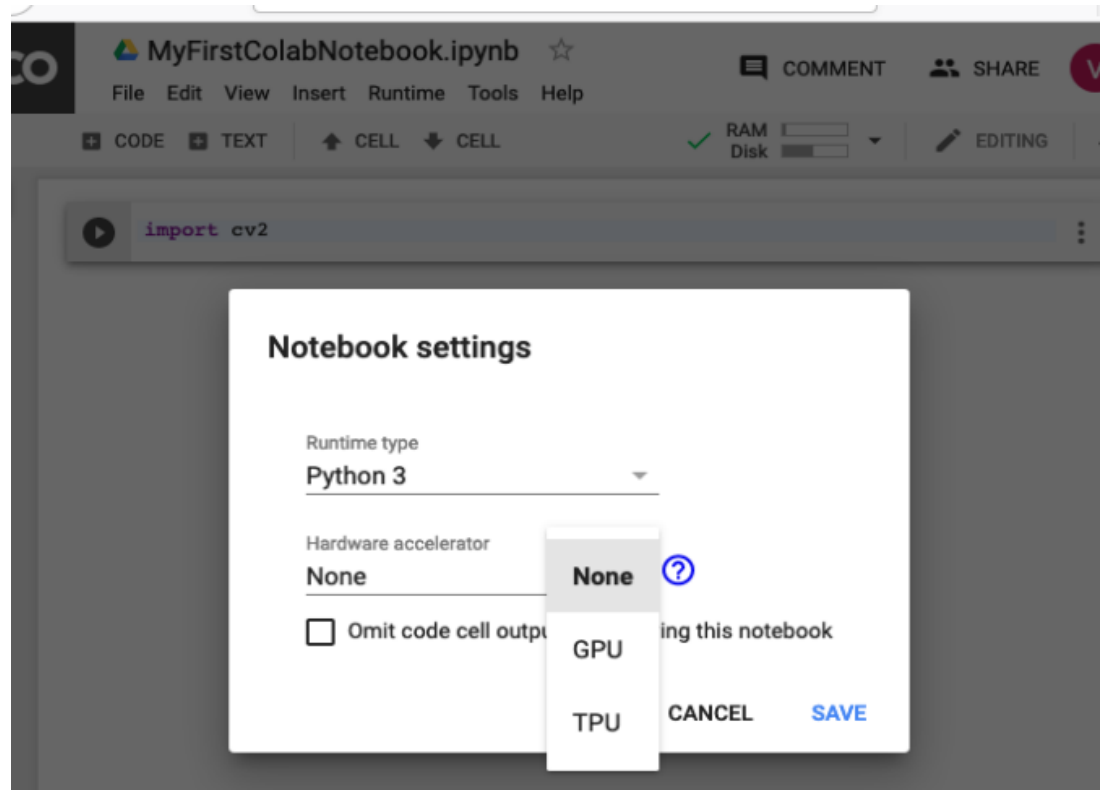
[Learn about the risks](#)

Cancel

Allow

Enabling GPU

- To enable GPU in your notebook, select the following menu options:
- Runtime / Change runtime type



Basic Python Example

Example:

- `print('Hello, World!')`
- `x = 10`
- `y = 5`
- `print(x + y)`

Working with Libraries

Example:

```
import numpy as np
```

```
arr = np.array([1,2,3])
```

```
print(arr * 2)
```

Data Handling Example

Example:

```
import pandas as pd
```

```
data = {'Name':['Ali','Sara'],  
        'Age':[20,22]}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

Visualization Example

Example:

```
import matplotlib.pyplot as plt
```

```
x = [1,2,3]
```

```
y = [2,4,6]
```

```
plt.plot(x,y)
```

```
plt.show()
```


Using Google Drive

```
from google.colab import  
drive
```

```
drive.mount('/content/drive')
```

Access your files directly from
Drive

Advantages & Limitations

Advantages:

Free GPU

Easy to use

Limitations:

Session timeout

Limited RAM

Conclusion

Google Colab is a powerful tool for beginners and researchers.

Perfect for AI, ML, and data science projects.

Lab 1: Python Basics for Artificial Intelligence

Instructor: Zeenat Zulfiqar
COMSATS University Islamabad Abbottabad Campus

Date: March 30, 2026

Contents

1	Objective	3
2	Part 1: Simple Python Programs for AI Basics	3
2.1	Activity 1: Display numbers on screen using Python IDLE . . .	3
2.2	Activity 2: Display strings on screen	3
2.3	Activity 3: Use Python as a calculator	3
2.4	Activity 4: Integer division and remainder	3
2.5	Activity 5: Calculate powers	4
2.6	Activity 6: Operator precedence expressions	4
2.7	Activity 7: Combine numbers and text	4
2.8	Activity 8: Take input from the keyboard	4
2.9	Activity 9: Check if a number is even (input from user)	4
2.10	Activity 10: Check if a number is even or odd (input from user)	5
2.11	Activity 11: Sum of all values from 0-10 using while loop . . .	5
2.12	Additional Exercises from previous Lab (Optional)	5
2.12.1	Hello World Program	5
2.12.2	Basic Input/Output	5
2.12.3	Arithmetic Operations	5
2.12.4	Odd or Even Checker	6
2.12.5	List of Squares	6
2.12.6	List Operations	6
2.12.7	Dictionary Operations	6
2.12.8	Function to Calculate Area of Circle	6
2.12.9	Lambda Function	6
2.12.10	List Comprehension for Even Numbers	7

3	Part 1: Simple Python Programs for AI Basics	7
3.1	Hello World Program	7
3.2	Basic Input/Output	7
3.3	Arithmetic Operations	7
3.4	Odd or Even Checker	7
3.5	List of Squares	8
3.6	List Operations	8
3.7	Dictionary Operations	8
3.8	Function to Calculate Area	8
3.9	Lambda Function	9
3.10	List Comprehension	9

1 Objective

To help students understand the basics of Python programming with a focus on concepts that are essential for Artificial Intelligence (AI).

2 Part 1: Simple Python Programs for AI Basics

2.1 Activity 1: Display numbers on screen using Python IDLE

```
1 # Display numbers
2 print(1)
3 print(123)
4 print(3.14)
```

2.2 Activity 2: Display strings on screen

```
1 # Display strings
2 print("Hello, AI students!")
3 print("Python is fun!")
```

2.3 Activity 3: Use Python as a calculator

```
1 # Python as a calculator
2 print("Sum:", 5 + 7)
3 print("Difference:", 10 - 3)
4 print("Product:", 4 * 6)
5 print("Quotient:", 12 / 3)
```

2.4 Activity 4: Integer division and remainder

```
1 # Integer division and remainder
2 a = 17
3 b = 4
4 print("Integer division:", a // b)
5 print("Remainder:", a % b)
```

2.5 Activity 5: Calculate powers

```
1 # Calculate powers
2 print("4^3 =", 4**3)
3 print("4^10 =", 4**10)
4 print("4^29 =", 4**29)
5 print("4^150 =", 4**150)
6 print("4^1000 =", 4**1000)
```

2.6 Activity 6: Operator precedence expressions

```
1 # Math expressions
2 expr1 = 2 + 3 * 4
3 expr2 = (2 + 3) * 4
4 expr3 = 10 / 2 + 6
5 expr4 = 2**3 + 5*2
6
7 print(expr1, expr2, expr3, expr4)
```

2.7 Activity 7: Combine numbers and text

```
1 # Combine numbers and text
2 age = 20
3 print("I am " + str(age) + " years old")
```

2.8 Activity 8: Take input from the keyboard

```
1 # Take input from user
2 name = input("Enter your name: ")
3 print("Welcome, " + name + "!")
```

2.9 Activity 9: Check if a number is even (input from user)

```
1 # Check if number is even
2 num = int(input("Enter an integer: "))
3 if num % 2 == 0:
4     print(num, "is even")
```

2.10 Activity 10: Check if a number is even or odd (input from user)

```
1 # Check if number is even or odd
2 num = int(input("Enter an integer: "))
3 if num % 2 == 0:
4     print(num, "is even")
5 else:
6     print(num, "is odd")
```

2.11 Activity 11: Sum of all values from 0-10 using while loop

```
1 # Sum using while loop
2 i = 0
3 total = 0
4 while i <= 10:
5     total += i
6     i += 1
7 print("Sum of numbers from 0 to 10:", total)
```

2.12 Additional Exercises from previous Lab (Optional)

2.12.1 Hello World Program

```
1 print("Hello, World!")
```

2.12.2 Basic Input/Output

```
1 name = input("Enter your name: ")
2 print("Hello, " + name + "!")
```

2.12.3 Arithmetic Operations

```
1 num1 = float(input("Enter first number: "))
2 num2 = float(input("Enter second number: "))
3 print("Sum:", num1 + num2)
4 print("Difference:", num1 - num2)
5 print("Product:", num1 * num2)
6 print("Quotient:", num1 / num2)
```


2.12.4 Odd or Even Checker

```
1 num = int(input("Enter a number: "))
2 if num % 2 == 0:
3     print(num, "is even")
4 else:
5     print(num, "is odd")
```

2.12.5 List of Squares

```
1 squares = [i**2 for i in range(1,11)]
2 print("Squares:", squares)
```

2.12.6 List Operations

```
1 numbers = [1,2,3,4,5]
2 numbers.append(6)
3 numbers.remove(3)
4 numbers.sort()
5 print("Updated list:", numbers)
```

2.12.7 Dictionary Operations

```
1 students = {"Ali":85, "Sara":92, "Ahmed":78}
2 students["Zara"] = 88
3 students["Ali"] = 90
4 del students["Ahmed"]
5 print("Updated students dictionary:", students)
```

2.12.8 Function to Calculate Area of Circle

```
1 import math
2 def area_of_circle(radius):
3     return math.pi * radius**2
4 r = float(input("Enter radius: "))
5 print("Area of circle:", area_of_circle(r))
```

2.12.9 Lambda Function

```
1 add = lambda x, y: x + y
2 print("Sum using lambda:", add(5,3))
```

2.12.10 List Comprehension for Even Numbers

```
1 even_numbers = [x for x in range(1,21) if x % 2 == 0]
2 print("Even numbers:", even_numbers)
```

3 Part 1: Simple Python Programs for AI Basics

3.1 Hello World Program

```
1 # Hello World Program
2 print("Hello, World!")
```

3.2 Basic Input/Output

```
1 # Take user's name and greet
2 name = input("Enter your name: ")
3 print("Hello, " + name + "!")
```

3.3 Arithmetic Operations

```
1 # Arithmetic operations
2 num1 = float(input("Enter first number: "))
3 num2 = float(input("Enter second number: "))
4
5 print("Sum:", num1 + num2)
6 print("Difference:", num1 - num2)
7 print("Product:", num1 * num2)
8 print("Quotient:", num1 / num2)
```

3.4 Odd or Even Checker

```
1 # Check if number is odd or even
2 num = int(input("Enter a number: "))
3 if num % 2 == 0:
4     print(num, "is even")
5 else:
6     print(num, "is odd")
```

3.5 List of Squares

```
1 # List of squares from 1 to 10
2 squares = []
3 for i in range(1, 11):
4     squares.append(i**2)
5 print("Squares:", squares)
```

3.6 List Operations

```
1 # List operations
2 numbers = [1, 2, 3, 4, 5]
3
4 # Add an element
5 numbers.append(6)
6
7 # Remove an element
8 numbers.remove(3)
9
10 # Sort the list
11 numbers.sort()
12
13 print("Updated list:", numbers)
```

3.7 Dictionary Operations

```
1 # Dictionary of students and marks
2 students = {"Ali": 85, "Sara": 92, "Ahmed": 78}
3
4 # Add a new student
5 students["Zara"] = 88
6
7 # Update marks
8 students["Ali"] = 90
9
10 # Delete a student
11 del students["Ahmed"]
12
13 print("Updated students dictionary:", students)
```

3.8 Function to Calculate Area

```
1 import math
2
```

```

3 # Function to calculate area of circle
4 def area_of_circle(radius):
5     return math.pi * radius**2
6
7 r = float(input("Enter radius: "))
8 print("Area of circle:", area_of_circle(r))

```

3.9 Lambda Function

```

1 # Lambda function to add two numbers
2 add = lambda x, y: x + y
3 print("Sum using lambda:", add(5, 3))

```

3.10 List Comprehension

```

1 # List of even numbers from 1 to 20 using list comprehension
2 even_numbers = [x for x in range(1, 21) if x % 2 == 0]
3 print("Even numbers:", even_numbers)

```

Graded Lab Tasks – Python Basics for AI

Instructor: Zeenat Zulfiqar
COMSATS University Islamabad Abbottabad Campus

Allotted Time: 1.5 Hours

Instructions

- Attempt all lab tasks individually.
- Use Python IDLE or any Python environment to implement the tasks.
- Submit your solutions on the CUI portal and provide a GitHub link for your code.

Lab Task 1: Student Grade Calculator

Write a Python program to accept marks of a student (1-100) and display the grade according to the following criteria:

- Grade F if marks are less than 50
- Grade E if marks are between 50 to 60
- Grade D if marks are between 61 to 70
- Grade C if marks are between 71 to 80
- Grade B if marks are between 81 to 90
- Grade A if marks are between 91 to 100

Lab Task 2: Factorial Calculator

Write a program that takes a number from the user and calculates the factorial of that number.

Lab Task 3: Fibonacci Series

Fibonacci series is defined as a sequence where the next number is formed by adding the previous two numbers. Start from 0 and 1. Example:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, ...

Steps:

1. Take an input number that shows how many terms to display.
2. Use loops to display the Fibonacci series up to that term.
3. Example: If the input number is 6, the output should be: 0, 1, 1, 2, 3, 5.